

gSpiceUI: Simulation Program With IC Emphasis

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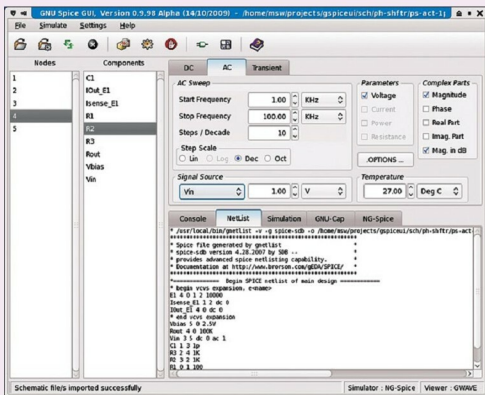
Tech geeks world over are most comfortable working at the back-end, creating hacks and tricks all the time. Like a field to a football player, they relish in the words put together that create magic on the computer screen. But there are those beginners, lost in the unfathomable ocean that is pages of codes and scripts. The source of hope to these newcomers is the graphical user interface (GUI).

In this month's DVD, we package gSpiceUI, a tool built with the intention of providing a GUI for electronic circuit simulation engines. The freely-available NG-Spice and GNU-Cap are the reasons behind why this tool was schemed up, and thanks to support from a few others, gSpiceUI has been up and running for over ten years today. With the above-mentioned software being independent of each other when it comes to the principle behind implementation, you as a user have the advantage of verifying your circuit using two simulation engines, thus moving a step closer towards reality.

gSpiceUI uses gnetlist tool to convert schematic files into tool-understandable netlist files, and Gwave or Graw for displaying the results from simulation, whereas gschem is the preferred tool for schematic capture—all of these being seamlessly integrated into this GUI master.

Built on a base of consistency

Simulating a circuit involves converting technical specifications into a language or structure that the software understands, working in sync with the simulation engine



The GNU Spice GUI

and, once completed, performing a stringent analysis with different types and methods. In all this, there is a high likelihood of your focus shifting from the primary target, the simulation itself. It is this factor that carved the foundation for gSpiceUI, and the focus is to abstract the technicalities from you, so as to let you focus entirely on the task at hand. Even while dealing with different analysis types and engines, the GUI is built to be as consistent as possible, reducing the hassles while simulating a circuit.

A smarttool that deals in netlists

gSpiceUI is smart and tries to parse simulation instructions, if any, in

the included netlist itself. The file's formatting is changed to suit the tool; whatever cannot be understood is removed. You might do better to send in a copy of the netlist to gSpiceUI, a file with lines well within the line limit set by you.

No matter how many independent schematics you import, gnetlist springs into action to combine these into a single netlist file, which is then taken forward successively. You can also choose to reload a given netlist or re-import from schematics at any stage of using this GUI tool.

Schematics can be edited with ease, but remember to load the files into the tool after the edit is complete. Just crosscheck the following minute details; else the simulation